Multiply up to four digits by two digits

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Calculate 2,304 x 27. Copy or complete the first step.

2. Calculate 4,216 x 34.

3. Calculate 3,075 x 46.

4. Miro says: Miro treats the 2 in 27 as 2 rather than 20. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate 2,304 x 27. Copy or complete the first step.
   Answer 62,208.

2. Calculate 4,216 x 34.
   Answer 143,344.

3. Calculate 3,075 x 46.
   Answer 141,450.

4. Miro says: Miro treats the 2 in 27 as 2 rather than 20. Is this correct? Explain.
   Answer The 2 represents two tens, so its partial product is twenty times the first factor.

Multiply up to four digits by two digits

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Calculate 4,216 x 34.

2. Calculate 3,075 x 46.

3. Use compensation to calculate 4,999 x 62.

4. Identify and correct this misconception: Miro treats the 2 in 27 as 2 rather than 20.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Calculate 4,216 x 34.
   Answer 143,344.

2. Calculate 3,075 x 46.
   Answer 141,450.

3. Use compensation to calculate 4,999 x 62.
   Answer 309,938.

4. Identify and correct this misconception: Miro treats the 2 in 27 as 2 rather than 20.
   Answer The 2 represents two tens, so its partial product is twenty times the first factor.

5. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Multiply up to four digits by two digits

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: _________________

1. Use compensation to calculate 4,999 x 62.

2. Explain why this reasoning fails: Miro treats the 2 in 27 as 2 rather than 20.

3. Create a new example that tests the same idea as: Calculate 3,075 x 46.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Use compensation to calculate 4,999 x 62.
   Answer 309,938.

2. Explain why this reasoning fails: Miro treats the 2 in 27 as 2 rather than 20.
   Answer The 2 represents two tens, so its partial product is twenty times the first factor.

3. Create a new example that tests the same idea as: Calculate 3,075 x 46.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.